Obsolete and Incidental Words Shift Language-Model Decisions, but Change Them Mainly When Another Cue Agrees

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Abstract

A model that reads an updated record can report the current value correctly and still let the old value shape its decision. We test when this happens, and whether it depends on the old value having been superseded. In synthetic routing and access-control records, the current state, the policy and the correct action are fixed while one obsolete state is edited to imply the correct or the wrong action. Matched controls place the same word as an updated assignment of an unrelated attribute, as a word the entity merely mentioned, or as an unattached word. In a frozen confirmation on held-out wording (Qwen3-8B) and in four further open models on identical rows, the obsolete state shifted the preference toward the action it implies (1.072–4.785 nats in routing) while the models retrieved the current and initial states correctly. These shifts changed decisions mainly when another cue also pointed to the wrong action. With the other entity's current state implying it, the net switch rate toward the wrong action was 0.143–0.233 in histories the models demonstrably track. In Qwen3-8B it fell to 0.031 when that state implied the correct action, and in both Qwen3 models to zero with thinking enabled. The interference does not require supersession: an incidental mention of the same word by the same entity shifted preferences in every model, and in Qwen3-8B changed more decisions than the superseded assignment in both designs.

1 Introduction

Agents and assistants increasingly act on records that change: a user's clearance is revoked, a ticket is rerouted, a preference is replaced. The record usually keeps the history, and the model is told that only the latest entry counts. Retrieval benchmarks ask whether the model can then report the current value, and recent work shows that old values do intrude on such reports, especially after many overwrites (Wang and Sun, 2026; Guo et al., 2026; Chattaraj and Raj, 2026). A deployed system, however, rarely asks the model to restate a value. It asks the model to *use* it. Two questions follow that retrieval accuracy does not answer. When a model can state the current value correctly, does the obsolete value still change the decision it takes, and under what conditions? And if it does, is the obsolete value acting as a superseded assignment of the queried attribute, or simply as a word that occurs in the context near the entity?

The second question matters because the two accounts call for different evaluations. If the interference is tied to supersession, it is a failure of state tracking, the target of update-aware memory, historical marking or attention routing toward the current entry (Guo et al., 2026). If any nearby mention of a decision-relevant word does the same, the problem is contextual entrainment (Niu et al., 2025) reaching decisions. We test no mitigation, but either way an evaluation of one needs a matched mention control and decision-level outcomes.

Design. We hold everything that should determine the decision fixed: the current state, the policy that maps states to actions, the query, and the correct action. We edit one obsolete state. In one member of each pair, the obsolete state implies the wrong action; in the other, it implies the correct action. The difference in the model's preference for the wrong action between the two members is the decision effect of the obsolete content, and the paired generated answers give its behavioral counterpart (Figure 1). The same edit is repeated in matched constructions that keep the word and its position but change its role: an assignment to a different attribute that was later updated, a word the entity only mentioned, and a word attached to no entity. Before testing decisions, we check on the same records that the model reports the current and the initial state correctly when asked.

Findings. The experiment ran under a protocol frozen before the confirmation data were generated,

on a held-out prompt template, with Qwen3-8B. Four further models (Qwen3-14B, Gemma 4 12B, Qwen3.5 9B, Granite 4.1 8B) were run on identical rows under a dated amendment; Gemma 3 4B and Ministral 3 14B did not meet the competence gate. Routing meets the competence criteria on the confirmation data in all five models; access does so only in Gemma 4 12B, and other access estimates are reported second and labeled. Four findings follow:

1. **Obsolete states shift decision scores in models that track them correctly.** In all five models, making the obsolete state imply the wrong action raised that action's log-odds (1.072–4.785 nats in routing), while the models retrieved the current and initial states correctly (§5.2, §5.6).

2. **The shifts change decisions mainly when another cue agrees.** In the main design the other entity's current state implies the wrong action. There, the net switch rate toward the wrong action was 0.143–0.233 across models in histories the models demonstrably track, almost never in reverse. In Qwen3-8B it fell to 0.031 when the other entity's state implied the correct action, and in both Qwen3 models to zero with thinking enabled, while preference shifts stayed (§5.5).

3. **Neither supersession nor assignment to the queried attribute is needed.** In all five models a superseded assignment interfered more in scores than an updated assignment to an unrelated attribute (0.311–3.102 nats), but in the two Qwen3 models the same assignment never updated was not resolved from it or interfered more. The same word mentioned in passing by the same entity shifted preferences in every model; it changed more decisions than the superseded assignment in Qwen3-8B (also without the second cue) and Granite 4.1 8B, and was not resolved from it in the other three (§5.3, §5.6).

4. **Easy-query sensitivity did not improve failure prediction** beyond the specified pre-decision features, in any model (§5.4).

What is and is not new. That stale context interferes with answers is established (Wang and Sun, 2026; Guo et al., 2026; Qian et al., 2026; Chao et al., 2026), as is the fact that mentioned tokens gain probability (Niu et al., 2025). Our contribution is a controlled test that connects the two at the level of decisions. We measure interference on policy decisions in open models, on items whose current and historical states the model retrieves correctly. We use minimal-pair edits that hold the correct action fixed, report both scores and generated answers, and include matched controls that separate an assignment to the queried attribute from an assignment to another attribute and from an incidental mention. We also report the negative result for a sensitivity-based predictor and, in Appendix A, two measurement problems we found along the way: chat-format events that the model never generates, and log-probabilities that change when tokens are appended under 8-bit and bfloat16 inference.

2 Related work

Interference from updated values. Proactive interference grows with the number of overwrites, and errors are mostly earlier values of the same key (Wang and Sun, 2026). Models retrieve the first value of a key more reliably than the latest (Chattaraj and Raj, 2026; Qiao et al., 2026). Guo et al. (2026) name the failure *stale binding*. They find that the current value stays linearly recoverable on failures, that historical wording reduces errors, and that attention routing repairs most of them. Their decision scenarios, with closed reasoning models, showed no difference between full histories and resolved snapshots. Benchmarks of forgetting and stale memory evaluate dialogue and agent settings by accuracy (Qian et al., 2026; Chao et al., 2026). STALE separates recognizing an update from applying it. We use a single update, decisions fixed by an explicit policy, and matched minimal pairs with lexical controls, below the interference loads at which retrieval breaks.

Entrainment and distraction. Tokens that appear in the context gain probability whether or not they are relevant (Niu et al., 2025), and irrelevant context degrades reasoning (Shi et al., 2023). Our mention and unattached-word constructions measure this in the update setting. Counterfactual edits are compound treatments that need matched alternative edits as baselines (Yang et al., 2026b), which is how our contrasts are built.

Binding, state tracking and conflicts. Mechanistic accounts of entity binding (Feng and Steinhardt, 2024; Gur-Arieh et al., 2026; Prakash et al.,

```
Read the dated record. Only the final assignment governs the decision.
Initially, EvbUnit0096A had routing state pending.        <- edited: pending | suspended
Initially, EvbUnit0096B had routing state suspended.
Later, EvbUnit0096A had archive tag restricted.
Finally, EvbUnit0096A had routing state restricted.
Later, EvbUnit0096B had archive tag restricted.
Finally, EvbUnit0096B had routing state pending.
Policy: state to bin: restricted -> B0, suspended -> B1, pending -> B2,
  active -> B3; bin to action: B0 -> EAST, B1 -> EAST, B2 -> WEST, B3 -> WEST
Apply the supplied rule to the final assignment of EvbUnit0096A; give its action.
Respond with exactly one label (EAST, WEST), with no other text.
```

Construction	Edited line (same word, same position)
S, superseded assignment	`Initially, EvbUnit0096A had routing state pending.`
O, other attribute, updated	`Initially, EvbUnit0096A had archive tag pending.`
M, mentioned by entity	`Initially, EvbUnit0096A mentioned the word pending.`
U, unattached word	`Initially, the unassigned word was pending.`

Figure 1: One routing confirmation item (the arrow is not part of the prompt). The current state `restricted` implies `EAST`; the edited word is `pending` (implies `WEST`, wrong) or `suspended` (correct). Below: the edited line in each construction.

2026) and state tracking (Tang et al., 2026; Kim and Schuster, 2023) may explain why lexical identity matters; we do not test them. Our records contain an intra-context conflict with an explicit temporal resolution (Xu et al., 2024). Questions about the value valid at a given time (Yang et al., 2026a) correspond to our historical retrieval check, not to our decision task.

3 Task, constructions and estimands

Records and decisions. Each history describes two entities. Each entity has an initial entry, an update of a separate *archive tag*, and a final assignment of the decision-relevant attribute (Figure 1). A policy maps four states to two actions, two states per action. In the *access* task, clearance maps directly to `APPROVE` or `DENY`. In the *routing* task, the routing state maps to a bin and the bin to `EAST` or `WEST`. The model is asked for the action implied by one entity's final assignment and must answer with exactly one label. Two vocabularies are used: natural states (`active`, `pending`, `restricted`, `suspended`) and opaque states (`S1`–`S4`). Natural labels carry prior meanings that the policy may contradict. Three orders define the cells: *sequential* (both initial entries, then both final blocks), *interleaved* (initial and final blocks alternate across entities), and *competing* (interleaved, plus two archived discussions that quote decision-relevant state words for the two entities). The factorial crosses task, cell, vocabulary, correct action, historical order, current order and entity orientation: 192 histories per replicate. State-to-label assignments rotate so that every state appears in every role. In every history the other entity's current state implies the wrong action, so confusing the two entities is always an error. The other entity's current state is fixed across the two members of a pair; interactions between the edited word and entity confusion remain possible.

Constructions. For the queried entity, the obsolete line takes one of four forms, all with the same edited word in the same slot: *superseded*, S (the queried attribute was initially assigned the word and later reassigned); *updated other attribute*, O (the archive tag was initially the word and was later updated); *entity mention*, M (the entity mentioned the word, with no assignment); and *unattached*, U (`the unassigned word was ...`). Every history also contains a *current-only* item with no initial lines, which measures whether the model can apply the policy at all, and a *live* item that edits the final assignment itself, which shows the size of a legitimate change. The constructions share the policy, query, archive-tag update, final assignments and surrounding text. They differ in syntax around the edited word, so the contrasts compare constructions as wholes, not isolated semantic features.

Decision effect. For a prompt x, let $L(x) = \log P(a_{\text{wrong}} \mid x) - \log P(a_{\text{correct}} \mid x)$, where each action event is the label exactly as the model writes it at the start of its turn, followed by the end-of-turn token. For construction c and history h, the two members differ only in the obsolete word: in member $x^{-}_{h,c}$ it implies the wrong action, and in

$x^+_{h,c}$ the correct one. The decision effect is

$$D_c = \frac{1}{|H|} \sum_{h \in H} \big[L(x^-_{h,c}) - L(x^+_{h,c}) \big]. \quad (1)$$

$D_c > 0$ means the obsolete content pushes preference toward the action it implies. It is a preference shift in nats, not an error rate. We write D_{S}, D_{O}, D_{M} and D_{U} for the four constructions, and $\Delta_{XY} = D_X - D_Y$ for a contrast. The two hypothesis contrasts are Δ_{SO}, which compares a superseded assignment to the queried attribute with an updated assignment to an unrelated attribute (both lines are later updated, so it does not isolate supersession itself), and Δ_{SM}, which asks whether assignment adds anything to an incidental mention.

Generated switches. With greedy decoding, a history scores $+1$ for construction c if the model answers wrongly under x^- and not under x^+, -1 for the reverse, and 0 otherwise. The mean over histories is the *net switch rate*, a value in $[-1, 1]$. The counts of histories with $+1$ and -1 are reported as *histories switched* to the wrong and to the correct action.

Retrieval checks. On each record we also ask *easy* queries for the current state of the queried entity and of the other entity. For superseded items we ask for the *initial* state, under an introduction that announces both initial and final assignments. The answers are generated and parsed strictly.

Easy-query sensitivity. Let w^- and w^+ be the two values of the edited word, and let $m_x(w)$ be the log probability that easy query q, asked on record x, assigns to the answer w (summed over its spaced, unspaced and capitalized surface forms, scored as answer prefixes). Define $E(q) = [m_{x^+}(w^+) - m_{x^+}(w^-)] - [m_{x^-}(w^+) - m_{x^-}(w^-)]$. Then $R_{\mathrm{easy}} = E(\text{queried entity}) - E(\text{other entity})$ is a per-history, query-specific sensitivity of retrieval scores to the edit. Hypothesis H3 asks whether R_{easy} helps predict downstream failures beyond features that are available before the decision query is asked.

4 Protocol

Model and runtime. Qwen3-8B (Yang et al., 2025) at a pinned revision (Appendix C), with thinking disabled through the chat template. Six further models were run later under dated amendments, with the same rows and rules (§5.6). Inference ran in float32 without quantization on one GPU. We chose float32 after finding that 8-bit and bfloat16 inference made the log probability at a fixed position change by up to 0.843 and 0.875 nats when tokens were appended after it. Float32 gave exactly zero (Appendix A.1). Action events were scored with a key–value-cache scorer. Before each run, the scorer was checked against one full forward pass per event on identical prompts (maximum difference 5.08×10^{-5} nats against a tolerance of 0.02). Answers were generated greedily with at most 16 new tokens and parsed strictly: an answer counts only if, after stripping outer whitespace, it is exactly one allowed label.

Splits and gate. Three disjoint splits were generated from separate seeds and separate entity-name pools. A development split (192 histories) was used only to check competence and answer formats. A frozen gate split (384 histories) decided eligibility. The confirmation split (768 histories, 8,448 records, 11 per history) used a held-out paraphrase of the introduction and query. A task-cell was *competent* if strict direct retrieval of the current state was at least 0.95, current-only decision accuracy at least 0.90, and invalid answers at most 0.05. Competence enters in three roles. (i) The frozen gate decided eligibility: confirmation data were generated only after the gate, recomputed from its saved scores, showed at least one competent cell. (ii) The replication amendment defined the cells behind each model's compact estimates as those competent on its frozen gate. (iii) Table 1, Table 3 and the text instead use the cells competent on the confirmation data, a documented departure from (ii) made after the confirmation results; Appendix B gives the estimates under (ii). All cells are reported. Task-level estimates average the fixed grid, and estimates from non-competent cells are labeled as such.

Hypotheses and statistics. The protocol fixed two primary hypotheses per task: H1, $D_{\mathrm{S}} > 0$; and H2, $\Delta_{\mathrm{SO}} \neq 0$, two-sided. Both use 97.5% percentile intervals from a history bootstrap (2,000 draws, fixed seed), Bonferroni-adjusted over the two tasks. Secondary outcomes use unadjusted 95% intervals: Δ_{SM}, the other construction effects, net switch rates, and H3. Histories are the resampling unit. Both members, all constructions and all queries of a history stay together. The two tasks are never pooled, and an interval that includes zero is reported as inconclusive, not as evidence of no effect.

Cell	Direct	No history	History	Competent
routing seq.	0.988	0.953	0.858	yes
routing int.	0.986	0.961	0.868	yes
routing comp.	0.982	0.953	0.846	yes
access seq.	0.916	0.867	0.831	no
access int.	0.945	0.875	0.858	no
access comp.	0.908	0.859	0.867	no

Table 1: Qwen3-8B, strict accuracy on the confirmation data. Direct: current state asked directly. No answer was invalid.

Prediction (H3). Predictors that are available before the decision query is asked: the easy query's margin between the current and the historical state, the historical candidate's log mass, easy confidence, the edited token's position, prompt length, and design indicators. The comparison adds R_{easy} to these. The outcome is a strict downstream failure on the historical constructions. Models are L2-regularized logistic regressions (scikit-learn; Pedregosa et al., 2011) with fixed settings, evaluated by history-grouped five-fold cross-validation; out-of-fold log loss with and without R_{easy} is compared by a paired history bootstrap (details in Appendix B). The downstream answer's own confidence is reported separately as a contemporaneous reference, not as a predictor, because it comes from the decision query itself.

Protocol history. The confirmation protocol is the second redesign of a design frozen before any model was run; every change, and the amendments that added models, are dated and listed in Appendix A. The records are internal, so we call them *frozen before confirmation*, not preregistered.

5 Results

All numbers in this section come from the frozen confirmation (Qwen3-8B, held-out template, 384 histories per task) unless marked otherwise. We give intervals only for the primary estimates; "not resolved" means that the interval includes zero. All intervals are in the tables of Appendix B.

5.1 Competence

On the confirmation data, the routing cells meet all competence criteria and the access cells do not, mainly because current-only decisions fall short of 0.90 (Table 1). Asked for the *initial* state on the superseded items, the model answered correctly in 0.990 of routing and 0.973 of access cases. In routing, then, the model reports the current state, reports the initial state, and applies the policy without history at or above 0.95. Access results are reported below as fixed-grid averages over cells that do not meet the competence criteria.

5.2 Obsolete states shift scores and decisions

H1 is supported in both tasks. In routing, when the superseded state implied the wrong action rather than the correct one, the wrong action's log-odds rose by $D_{\text{S}} = 3.133$ nats (97.5% interval $[2.743, 3.516]$). In access, where cells do not meet the competence criteria, the shift was 3.904. For scale, editing the *current* state itself, which legitimately changes the correct action, moved the same quantity by -8.197 (routing) and -10.100 nats (access).

The shift reaches generated answers. In routing, the net switch rate toward the wrong action was 0.185 $[0.143, 0.227]$: 75 histories switched to the wrong action and 4 to the correct one. In each routing cell, it was between 0.164 and 0.227 (Table 5). In access it was 0.167 (65 against 1). Figure 4 in Appendix B shows the same effect as accuracy. In every routing cell, decisions were correct in 0.906–0.953 of histories when the superseded state implied the correct action, and in 0.727–0.750 when it implied the wrong one.

Histories the model demonstrably tracks. Aggregate retrieval accuracy does not show that the switching histories are ones the model reads correctly. In a post hoc analysis, defined after the Qwen3-8B confirmation results and applied unchanged to Qwen3-14B, we kept a history only if three conditions held on the same records: the model retrieved the current state correctly under both members of the pair, retrieved the initial state correctly under both superseded members, and decided the current-only item correctly. This kept 348 of 384 routing histories for the superseded construction (Table 4 in Appendix B). In them, the net switch rate toward the wrong action was 0.158 (58 histories to the wrong action, 3 to the correct one) and the preference shift 2.990 nats. The effect is therefore not carried by histories in which the model misreads the record. In access, the same restriction keeps 284 of 384 histories and gives a net switch rate of 0.176 (50 against 0). In every one of these histories, though, the other entity's current state also implies the wrong action; §5.5 removes that cue.

Constr.	Net switch	W/C	Pair	Difference
S	+0.161	57/3	M − S	+0.104
O	+0.045	27/12	S − O	+0.116
M	+0.266	89/0	O − U	+0.012
U	+0.033	16/5		

Table 2: Qwen3-8B, routing: the 335 of 384 histories tracked under all four constructions (post hoc). W/C: switched to wrong / correct. Grey: interval includes zero (Table 12).

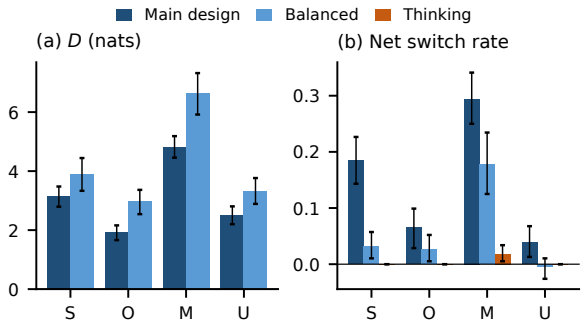


Figure 2: Qwen3-8B, routing, per construction, with 95% intervals. Thinking mode has no scores.

5.3 Attribute and mention contrasts

This section reports Qwen3-8B; §5.6 gives the other models. H2 compares the superseded assignment with an assignment of the same word to a different attribute that was later updated. It is supported in routing, $\Delta_{\mathrm{SO}} = +1.224$ (97.5% interval $[0.973, 1.485]$), with generated switches agreeing (paired difference $+0.120$). In routing, then, a superseded assignment to the queried attribute produces more interference than an updated assignment to an unrelated attribute. In access (non-competent cells) neither the contrast ($+0.035$) nor its switch difference is resolved.

The mention control, a secondary outcome, shows that supersession is not the largest source of pressure in Qwen3-8B. When the same entity only *mentioned* the word, the effect was larger than for the superseded assignment. In routing, $D_{\mathrm{M}} = 4.807$ against 3.133, so $\Delta_{\mathrm{SM}} = -1.675$. Net switch rates were 0.294 (113 to wrong, 0 to correct) against 0.185, a paired difference (M − S) of $+0.109$. In access, $D_{\mathrm{M}} = 5.182$ against 3.904 ($\Delta_{\mathrm{SM}} = -1.278$); the switch difference there is not resolved. A word attached to no entity still shifted preference substantially, by 2.496 (routing) and 3.043 nats (access), but rarely changed a generated decision (0.039 and 0.062). On the routing histories tracked under all four constructions (Table 2; post hoc), the mention exceeded the superseded assignment in net switch rate, and the superseded assignment exceeded the updated unrelated attribute; the unrelated attribute and the unattached word were not resolved.

An other attribute that is never updated. A fifth construction, N, added by a dated amendment before its data were scored, removes the later update of the other attribute (a filler line of the same length replaces it) on the 384 routing confirmation histories of both Qwen3 models (Table 9 in Appendix B). As predicted, the update reduced interference ($\Delta_{\mathrm{ON}} = -1.204$ and -1.942 nats). Never updated, the other attribute was not resolved from the superseded assignment in Qwen3-8B ($\Delta_{\mathrm{SN}} = +0.020$, switches -0.008) and interfered more in Qwen3-14B (-1.076, switches -0.096). Δ_{SO} thus compares the queried attribute with one whose value was later replaced; it does not show that assignment to the queried attribute matters.

Across four earlier, non-confirmatory splits, Δ_{SM} never favored the superseded assignment, but its size varied with the template (Appendix B, Figure 5).

5.4 Easy-query sensitivity did not improve failure prediction

The edit strongly affected easy retrieval scores: the mean R_{easy} on superseded items was 8.096 nats in access and 10.878 in routing. That sensitivity did not improve prediction beyond the specified pre-decision features under the specified logistic model (Table 7 in Appendix B). Pre-decision features reduced log loss from 0.414 to 0.346 and reached AUROC 0.780. Adding R_{easy} reduced log loss by 0.0004 nats (not resolved); the upper end of its interval is under 3% of what the simple features already gain. The result is the same in each task, for the wrong-answer outcome, and in every further model.

5.5 When score shifts change decisions

Two further runs on Qwen3-8B ask when the score shift becomes a different decision (Figure 2; Table 8). In the main design, confusing the two entities gives the wrong action, because the other entity's current state implies it. A balanced subset, recorded before its data were scored, sets the other entity's current state to the queried entity's, so that

it implies the correct action (192 new routing histories, held-out template; Appendix A). The model retrieved both states and decided correctly without history in every balanced history. D_S stayed positive (3.895), but the net switch rate fell to 0.031 (6 histories to wrong, none to correct), against 0.185 in the main design. The mention kept a net switch rate of 0.177 (34 against 0). In this model, then, the superseded assignment shifts scores with or without the second cue, but mostly changes decisions when that cue also points to the wrong action; the mention depends less on it. The balanced histories differ from the main ones, and identical current states may make the decision easier, so the comparison is descriptive.

Second, we generated answers with thinking enabled on the main-design routing rows (greedy, at most 4,096 new tokens; no scores). In Qwen3-8B, current-only accuracy was 1.000 and 2 of 3,456 items were invalid. No history switched for the superseded, other-attribute or unattached constructions; the mention switched 7 to the wrong action (0.018). In Qwen3-14B (current-only accuracy 1.000, none invalid), only 2 other-attribute and 1 mention histories switched, all to the wrong action. Thinking and direct answers are different generation processes, and no test was prespecified.

5.6 Further models

Four further models were run under amendments dated before any of their data were scored, with the same rows, rules and runtime (Appendix A): Qwen3-14B, then Gemma 4 12B, Qwen3.5 9B and Granite 4.1 8B. Table 3 gives each on the cells competent on the confirmation data: all three routing cells for Qwen3-14B, two for the others, and access only for Gemma 4 12B.

The obsolete-state effect replicates in every model. In routing, D_S is 1.072–4.785 nats, and the net switch rate in tracked histories is 0.143–0.233, with at most 3 histories switching to the correct action (Table 11). Δ_{SO} is positive in every model; its paired switch difference is resolved in Gemma 4 12B and Qwen3.5 9B, not in Qwen3-14B or Granite 4.1 8B. The mention comparison splits. In Granite 4.1 8B, as in Qwen3-8B, the mention moved scores and decisions more than the superseded assignment. In Gemma 4 12B the superseded assignment moved scores more ($\Delta_{SM} = +1.513$), with no resolved switch difference. In Qwen3-14B and Qwen3.5 9B neither difference is resolved; Qwen3.5 9B shows the smallest score shifts and the largest switch rate. We do not read the differences between models as a trend with size or family. Further Qwen3-14B estimates, and estimates under the gate rule, are in Appendix B.

Gemma 3 4B and Ministral 3 14B met no gate cell: current-only decision accuracy was 0.609–0.812 and direct retrieval 0.916–0.999 (Table 10), so no confirmation split was generated. These are eligibility results, not evidence for or against stale-state interference in these models.

6 Discussion

What the evidence supports. In five models that report both the current and the initial state correctly, an obsolete state still shifts the preference between routing actions toward the action it implies. Whether that shift changes the decision depends on the rest of the record. When the other entity's current state also implies the wrong action, the net switch rate is 0.143–0.233 in histories the models demonstrably track; in Qwen3-8B, without that cue it is 0.031, and with a reasoning trace it is zero. A plausible reading is that the obsolete word adds pressure that crosses the decision boundary mostly when a second cue already pushes toward the same action. Stale-value interference, documented for value queries (Guo et al., 2026; Wang and Sun, 2026), thus reaches single-step policy decisions after a single update, on items where retrieval is at ceiling, but conditionally. The zero switch rate with thinking agrees with Guo et al. (2026), whose reasoning models showed no difference between full histories and resolved snapshots. The interference does not require supersession, nor assignment to the queried attribute: an incidental mention of the same word shifts preferences in every model, and in Qwen3-8B it changed decisions even without the second cue (0.177) and, a few times, with thinking. It behaves like entrainment by decision-relevant words attached to the queried entity (Niu et al., 2025); how it compares with the superseded assignment differs between models. An unattached word likewise shifted preferences by 2.5–3.0 nats but changed few decisions (0.039 and 0.062): large changes in scored preference need not change greedy generation.

Consequences for evaluation and mitigation. Three practical points follow. First, stale-state benchmarks that vary only update histories cannot tell supersession from incidental mention; a matched mention condition is cheap and changed

Model	Cells	n	Scores (nats)			Net switch rate		
			D_S	Δ_{SO}	Δ_{SM}	S	S − O	M − S
Qwen3-8B	routing comp., int., seq.	384	+3.133	+1.224	−1.675	+0.185	+0.120	+0.109
Qwen3-14B	routing comp., int., seq.	384	+4.785	+0.866	+0.248	+0.167	+0.021	+0.003
Gemma 4 12B	routing comp., seq.	256	+3.709	+3.102	+1.513	+0.109	+0.090	−0.012
Gemma 4 12B	access comp., int., seq.	384	+4.285	+1.561	+1.455	+0.083	−0.031	−0.010
Qwen3.5 9B	routing int., seq.	256	+1.072	+0.311	−0.019	+0.242	+0.148	−0.062
Granite 4.1 8B	routing comp., int.	256	+3.385	+0.755	−1.929	+0.188	+0.027	+0.156

Table 3: Confirmation results on each model's competent cells (access only for Gemma 4 12B). Grey: interval includes zero (intervals in Table 13).

our conclusion. Second, decision-level rates depend on what else the record says, so a benchmark should report which other cues agree with the stale value. Third, mitigations (which we did not test) should be evaluated the same way, with and without a reasoning step, and should also check historical retrieval, which a blunt fix could damage.

Alternative explanations. The constructions differ in syntax around the edited word, so the contrasts compare whole constructions. Correct direct retrieval also does not show that the current state was the one used at decision time, so the retrieval checks do not rule out stale binding in the sense of Guo et al. (2026). The mention construction could be read as more salient. Its advantage varied with template in Qwen3-8B, was not resolved in three models and reversed in scores in Gemma 4 12B, which fits a surface explanation better than a categorical one. Natural state words carry prior meanings, but in Qwen3-8B D_S was positive and Δ_{SM} negative in all 12 task × cell × vocabulary strata (Table 6).

7 Conclusion

In five open models that report the current and the initial state of a record correctly, an obsolete state still shifts routing preferences toward the action it implies. Those shifts changed decisions mainly when another cue in the record agreed: net switch rates were 0.143–0.233 when the other entity's current state implied the wrong action, in Qwen3-8B 0.031 when it did not, and in both Qwen3 models zero with thinking enabled. The interference requires neither supersession nor assignment to the queried attribute: the same word mentioned in passing by the same entity shifted preferences in every model and changed more decisions than the superseded assignment in two. Retrieval-score sensitivity did not improve prediction beyond the specified pre-decision features. Stale-context benchmarks and mitigations should therefore be evaluated with incidental-mention controls, with the other cues in the record reported or varied, and with decision-level outcomes, not only retrieval accuracy. Two further models could not perform the base task reliably enough to be evaluated.

Limitations

Five models, three developers. The confirmed results are for Qwen3-8B, Qwen3-14B, Gemma 4 12B, Qwen3.5 9B and Granite 4.1 8B, all open models of 8–14B parameters run with thinking disabled. Gemma 3 4B and Ministral 3 14B did not meet the competence gate. The never-updated control was run only in the two Qwen3 models, the balanced subset only in Qwen3-8B and thinking mode only in the Qwen3 models, so the conditional decision effect rests on one model. The mention comparison differs between models. We make no claim about larger or closed models.

Synthetic tasks. The records, policies and constructions are synthetic and short: two entities, one update and an explicit policy. This control is what lets the contrasts hold the correct action fixed, but the effect sizes need not transfer to natural dialogue, long agent logs or tool outputs.

Competence in access. The access cells met the competence criteria on the confirmation data only in Gemma 4 12B. In the Qwen3 models this is mainly because current-only accuracy was 0.859–0.875 for Qwen3-8B and 0.820–0.859 for Qwen3-14B. The errors differ by model. Qwen3-8B errs mostly with natural state words (current-only accuracy 0.797, against 0.938 with opaque labels), consistent with prior word meanings competing with the policy. Qwen3-14B errs mostly when the correct action is `DENY` (0.688, against 0.990 for `APPROVE`), with opaque labels as well (0.615), which points to a response bias rather than a word

prior. Access estimates are fixed-grid averages over cells where the model already makes errors without history.

Construction differences. Constructions share the edited word and its position but differ in syntax. The contrasts compare constructions, not isolated semantic properties. The template dependence of the mention contrast shows that wording matters.

Behavioral evidence only. We measure scores and generated answers. Nothing here identifies a mechanism, and terms such as *entrainment* describe the behavioral pattern, not a verified internal process.

Prediction scope. H3 uses cross-validation within the confirmation split. A null result there does not rule out other score-based diagnostics, other outcomes, or other models.

No mitigation. We did not test an intervention; the paper makes no claim about which mitigations work.

Use of AI assistants. A large language model assisted with implementing the experiment code and drafting this manuscript. The authors checked the claims, analyses, references and final text.

Ethical Considerations

All data are synthetic records generated from templates. No human subjects, annotators or personal data were involved. The results bear on systems that act on changing records, such as access control and routing. They show that an incidental mention near an entity can shift an automated decision. This could be used deliberately: someone able to insert text into a record could bias a model-mediated decision without changing any authoritative value. We report the effect so that evaluations and safeguards can account for it. Our experiments do not test any deployed system. The models were used for research under their published licenses (Appendix C).

References

H. Chao, Y. Bai, R. Sheng, T. Li, and Y. Sun. 2026. STALE: Can LLM agents know when their memories are no longer valid? *arXiv preprint arXiv:2605.06527*.

S. Chattaraj and K. Raj. 2026. LLMs remember first, forget last: Dual-process interference in large language models. *arXiv preprint arXiv:2603.00270*.

T. Dettmers, M. Lewis, Y. Belkada, and L. Zettlemoyer. 2022. LLM.int8(): 8-bit matrix multiplication for transformers at scale. In *Advances in Neural Information Processing Systems (NeurIPS)*.

J. Feng and J. Steinhardt. 2024. How do language models bind entities in context? In *International Conference on Learning Representations (ICLR)*, pages 36391–36413.

J. Guo, Y. Fang, S. Gu, C. Spanos, J. Demmel, and J. Lavaei. 2026. When context changes: Understanding update failures in LLMs. *arXiv preprint arXiv:2609.38866*.

Y. Gur-Arieh, M. Geva, and A. Geiger. 2026. Mixing mechanisms: How language models retrieve bound entities in-context. In *International Conference on Learning Representations (ICLR)*, pages 27757–27786.

N. Kim and S. Schuster. 2023. Entity tracking in language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3835–3855.

J. Niu, X. Yuan, T. Wang, H. Saghir, and A. H. Abdi. 2025. Llama see, llama do: A mechanistic perspective on contextual entrainment and distraction in LLMs. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16218–16239.

A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Köpf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, and 2 others. 2019. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32.

F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. 2011. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830.

N. Prakash, N. Shapira, A. Sen Sharma, C. Riedl, Y. Belinkov, T. Rott Shaham, D. Bau, and A. Geiger. 2026. Language models use lookbacks to track beliefs. In *International Conference on Learning Representations (ICLR)*, pages 7657–7694.

Y. Qian, Z. Yang, W. Ding, H. Li, and Y. Xie. 2026. Do LLMs forget what they should? evaluating in-context forgetting in large language models. In *International Conference on Learning Representations (ICLR)*, pages 108956–108983.

B. Qiao, S. Guo, X. Yang, K. Li, W. Zhou, S. Hu, and Y. Song. 2026. Diagnosing retrieval bias under multiple in-context knowledge updates in large language models. *arXiv preprint arXiv:2603.12271*.

F. Shi, X. Chen, K. Misra, N. Scales, D. Dohan, E. H. Chi, N. Schärli, and D. Zhou. 2023. Large language models can be easily distracted by irrelevant context. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *Proceedings of Machine Learning Research*, pages 31210–31227.

Z. Tang, Q. Zhao, G. Franco, D. T. Wijaya, A. Mueller, S. Schuster, and N. Kim. 2026. Do language models track entities across state changes? In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 119466–119503.

C. Wang and J. V. Sun. 2026. Unable to forget: Proactive interference reveals working memory limits in LLMs beyond context length. In *Conference on Language Modeling (COLM)*.

T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz, J. Davison, S. Shleifer, P. von Platen, C. Ma, Y. Jernite, J. Plu, C. Xu, T. Le Scao, S. Gugger, and 3 others. 2020. Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45.

R. Xu, Z. Qi, Z. Guo, C. Wang, H. Wang, Y. Zhang, and W. Xu. 2024. Knowledge conflicts for LLMs: A survey. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 8541–8565.

A. Yang et al. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

X. Yang, Z. Liu, R. Li, and Y. Lei. 2026a. Large language models in resolving contextual knowledge conflicts. *arXiv preprint arXiv:2609.03148*.

Z. Yang, M. Levy, Y. Goldberg, and B. C. Wallace. 2026b. Compared to what? baselines and metrics for counterfactual prompting. In *Conference on Language Modeling (COLM)*.

A Protocol history

Every change below is recorded, with its date, in the protocol documents released with the code. Each paragraph states what the change preceded. The Qwen3-8B protocol was fixed before its confirmation data were generated. The second-model and replication amendments were written after Qwen3-8B's confirmation results, but before any data from the models they added. Two analysis choices were made after confirmation results: the tracked-history analysis (after Qwen3-8B's) and the confirmation-data rule for competent cells (after Qwen3-14B's).

v1 (frozen design, never run; audited 2026-10-08). The design fixed the hypotheses, estimands, factorial, constructions, gates, bootstrap and held-out confirmation template. An audit before any inference found two problems. (i) The action events began with a space (␣APPROVE). Qwen3-8B begins its turn without one, and the two strings tokenize differently, so v1 would have scored continuations the model never produces. (ii) Among the H3 predictors was the decision's own action confidence, which is not available before the decision is made.

v1b (2026-10-08, before any inference). Action events became the label exactly as generated plus the end-of-turn token. The H3 predictors were split into pre-decision predictors and a contemporaneous reference. Historical retrieval was added as a competence measure. During setup, the runtime was changed twice for numerical reasons (Appendix A.1), and a rounding tolerance of 10^{-5} was added to validators that rejected float32 log probabilities a few 10^{-7} above zero. The v1b gate failed: direct retrieval was 0.851–0.946 in every cell. The cause was the prompt format. The user message ended with a literal `Answer:` line, which the model often echoed (about 260 of 4,224 retrieval answers); about 100 answers gave an action label instead of a state. Only 64 named the obsolete value. The decision effects replicated on the gate histories.

v2 (2026-10-09, before any v2 data; first redesign, at most two were allowed). The redesign changed only what the failure required: it removed the trailing `Answer:` line and used new seeds and entity-name pools so that all histories were new. Confirmation size was set to four replicates (768 histories) before any v2 data existed, to fit the compute budget. The scorer was changed from a batched scorer to a key–value-cache scorer for speed, before any v2 effect was inspected. Both scorers compute the same quantity, and the cache scorer passed its equivalence check. The v2 gate passed in three cells, and confirmation data were then generated.

Second model (2026-10-09, after Qwen3-8B's confirmation results, before any Gemma inference). The model list, adding Gemma 3 4B with the same rows, rules and runtime, was fixed before any Gemma data were scored. A 24-record smoke test, inspected for format only, showed that Gemma writes its label, a newline and then its end-of-turn token. The scored events had omitted the newline

and covered at most a few percent of the probability mass. The scored event was corrected to include the newline, the checks were rerun, and the smoke outputs were set aside. Development and gate then failed the competence criteria in every cell (§5.6).

Replication models (2026-10-09, after Qwen3-8B's confirmation results, before any inference for the models listed). A further amendment listed five models in a fixed order with pinned revisions. It fixed the per-model sequence, the competence criteria, the reporting rule, and the definition of competent cells: those passing on the model's own frozen gate. For this submission the scope was reduced, before any of the five had produced data, to the first model, Qwen3-14B. The other four are deferred under the same amendment. Qwen3-14B's preflight needed no correction (causality 0.0; scorer difference 8.3×10^{-7} nats; it ends its answer with the label and its end-of-turn token). Its rows were byte-identical to Qwen3-8B's in every split.

Models from further families (2026-10-10, after the Qwen3 confirmation results, before any data from the models listed). A separate amendment listed Ministral 3 14B, Gemma 4 12B, Qwen3.5 9B, Granite 4.1 8B and OLMo 3 7B, in that order and at pinned revisions, with the same rows, rules, competence criteria and runtime. A preflight causality check had to give exactly 0.0. It did for the first four; OLMo 3 gave 2.8×10^{-5} nats and was not run. Two faults stopped the smoke stage and were corrected before any development data: the format check summed the retrieval candidates' keys instead of their values, and Gemma 4's generation configuration sets two token scores to $-\infty$, which tripped a finiteness check on processed scores (the check now reads raw logits; decoding and scores are unchanged). Ministral 3 14B met no gate cell and was not confirmed. The other three produced rows byte-identical to Qwen3-8B's in every split.

Analysis choices after confirmation results. The tracked-history analysis (Tables 4, 2 and 11) was defined on 2026-10-09, after Qwen3-8B's confirmation results and before any Qwen3-14B inference, and is post hoc. Its common-set version (Tables 2 and 12) was added on 2026-10-10, after both models' results. Reporting competent cells by the confirmation data rather than by each model's frozen gate was decided on 2026-10-10, after Qwen3-14B's confirmation results; Appendix B gives the gate-rule estimates.

Never-updated other attribute (2026-10-10, after both models' confirmation results, before any of its data). A dated amendment added the fifth construction of Table 9, its contrast Δ_{ON} and the prediction $\Delta_{\mathrm{ON}} \leq 0$, using the sealed routing confirmation histories of Qwen3-8B and Qwen3-14B and the same runtime. Before scoring, each model passed the scorer and causality checks (causality exactly 0.0). As a reproducibility check, 96 records of the original other-attribute construction were rescored; their action scores matched the sealed ones exactly and all generations were identical.

Balanced subset (2026-10-10, after the Qwen3 confirmation results, before any of its data). The same amendment generated 192 new routing histories (seed 82101, held-out template, a disjoint name pool) in which the other entity's current state equals the queried entity's, so that it implies the correct action; everything else follows the confirmation design. Qwen3-8B was scored with the confirmation runtime. The amendment asked only whether D_{S} and the superseded net switch rate stay positive; no test was prespecified. Its causality check names no threshold. Of 24 checks, 23 gave exactly 0.0 and one gave 5.7×10^{-5} nats, within the runner's tolerance; we report the subset under that rule. The further-families amendment, by contrast, required exactly 0.0 and excluded OLMo 3 at 2.8×10^{-5}.

Thinking mode (2026-10-10, before any thinking-mode data). A separate amendment generated answers with thinking enabled for Qwen3-8B and Qwen3-14B on the sealed routing confirmation rows (superseded, other-attribute, mention and unattached members, and current-only items; 3,456 items per model), greedy, at most 4,096 new tokens, with the strict parser applied to the text after the last closing think marker. Truncated, unclosed or unparseable answers are invalid and count as not wrong. A smoke run fixed the item set by projected cost (all items for both models). Ministral 3 Reasoning was listed conditional on its instruction-tuned model passing the gate, which it did not. Qwen3-8B's 2 invalid items had traces that did not close within the limit; median traces were 525 (Qwen3-8B) and 404 tokens (Qwen3-14B).

A.1 Numerical causality of log-probability scoring

Scoring answers by teacher forcing assumes that the log probability at a position does not depend on tokens appended after it. We tested this directly on the same prompts, by comparing the next-token distribution at a fixed position with and without one or two tokens appended. With 8-bit weights (LLM.int8(), Dettmers et al., 2022), log probabilities changed by up to 0.843 nats. With bfloat16, the top-20 log probabilities changed by up to 0.875 nats, and computing only the output layer in float32 reduced it to 0.502. In float32, and with 4-bit nf4 weights, it was exactly zero. In practice this broke our validators: four disjoint candidate answers summed to more than probability one. A key–value-cache scorer also disagreed with full recomputation by 3.406 nats under 8-bit weights, but by 5.08×10^{-5} in float32. Score differences of a few nats are the size of the effects studied here, so we ran all inference in float32. Score-based studies that use reduced precision should report a check of this kind.

B Additional results

This appendix gives the decision effects by construction for both tasks (Figure 3), retrieval and decision accuracy per cell (Figure 4), the tracked-history analysis per cell (Table 4), the per-cell switch rates behind §5.2 (Table 5), the H3 prediction table (Table 7), and the per-stratum decision effects referred to in §6 (Table 6). It also shows the replication of the primary and mention contrasts across all splits (Figure 5, summarized in §5.3), the gate results of every model attempted (Table 10), further Qwen3-14B estimates and estimates under the gate rule, the tracked-history analysis per model (Tables 11 and 12), and Qwen3-14B's construction effects (Figure 6). Rows are ordered with routing, the task competent on the confirmation data, first.

Tracked histories. Current-state retrieval is checked per construction, so retained sets differ slightly across constructions. Initial-state retrieval is asked only on superseded members and is applied at the history level to every construction.

Replication across splits. Four earlier, non-confirmatory splits of the same protocol family (development and gate of the first redesign and of the confirmed protocol, on the standard template) show which conclusions depend on wording (Figure 5). D_S was positive in all ten split-by-task estimates (3.133 to 6.599 nats), and Δ_{SO} in nine of ten. Δ_{SM} never favored the superseded assignment. On the standard template, however, it ranged from -0.806 to $+0.163$, with six of eight intervals including zero. The size of the mention advantage depends on wording.

	Kept	D_S	Net switch rate	Switched
routing seq.	118/128	2.959	0.144 [0.076, 0.220]	19 / 2
routing int.	114/128	3.252	0.202 [0.132, 0.281]	24 / 1
routing comp.	116/128	2.764	0.129 [0.069, 0.190]	15 / 0
routing, all	348/384	2.990	0.158 [0.118, 0.198]	58 / 3
access, all	284/384	4.059	0.176 [0.134, 0.218]	50 / 0

Table 4: Qwen3-8B, histories the model tracks (§5.2), with 95% intervals. Kept: retained of those in the cell. Switched: to wrong / correct.

Cell	Superseded	Other attr.	Mention	Unattached
routing seq.	0.164	0.062	0.352	0.055
routing int.	0.227	0.078	0.273	0.055
routing comp.	0.164	0.055	0.258	0.008
access seq.	0.133	0.070	0.148	0.094
access int.	0.273	0.234	0.219	0.094
access comp.	0.094	0.109	0.219	0.000

Table 5: Net switch rate toward the wrong action, per cell and construction (confirmation, $n = 128$ histories per cell). Intervals are in the released analysis.

H3 method. Each fold's model is a pipeline of feature standardization (zero mean, unit variance) followed by an L2-regularized logistic regression ($C = 1$, lbfgs, no tuning). Both steps are fitted on the training folds only, so the held-out fold never informs the scaling. Folds are grouped by history (five folds). For each history we average, over its items, the out-of-fold log loss without minus with R_{easy}; the reported gain and its 95% interval come from the history bootstrap of these per-history differences (2,000 draws, fixed seed).

Qwen3-14B details. Its frozen gate passed in all three routing cells and in no access cell. On the confirmation, routing retrieval was 0.989–0.997 for the current state and 0.997 for the initial state, and current-only decisions were 0.977–0.992 correct. In routing, the net switch rate is 0.167 over all histories (67 switched to wrong, 3 to correct) and 0.165 $[0.128, 0.205]$ in the 376 tracked histories. In generated switches the attribute difference is not resolved (0.167 against 0.146, paired difference $+0.021$ $[-0.023, +0.068]$). The mention and the superseded assignment had similar switch rates:

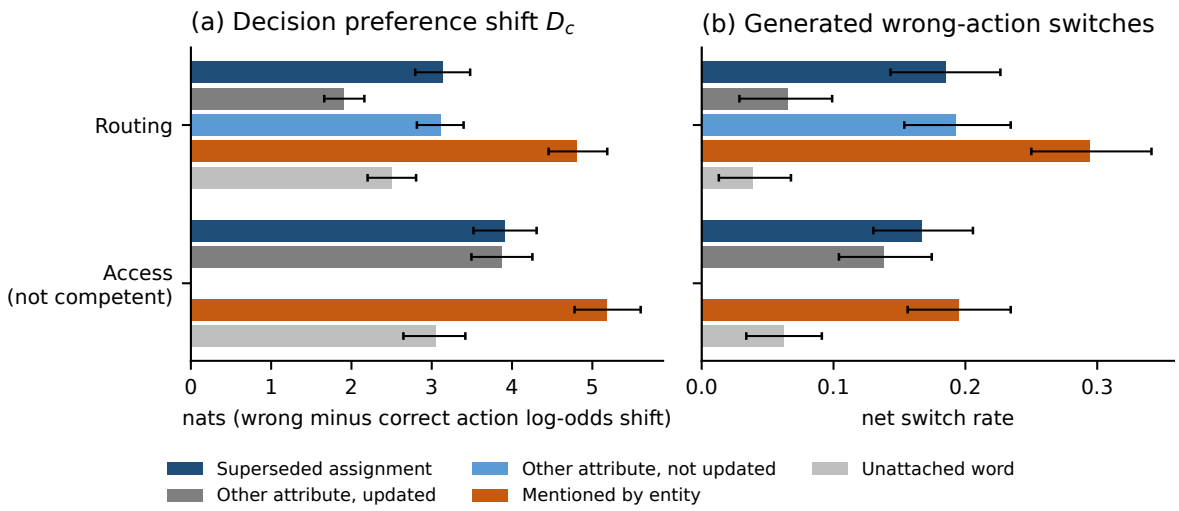


Figure 3: Qwen3-8B, effects by construction with 95% intervals: (a) D_c, (b) net switch rate. Access cells are not competent; N was scored on routing only.

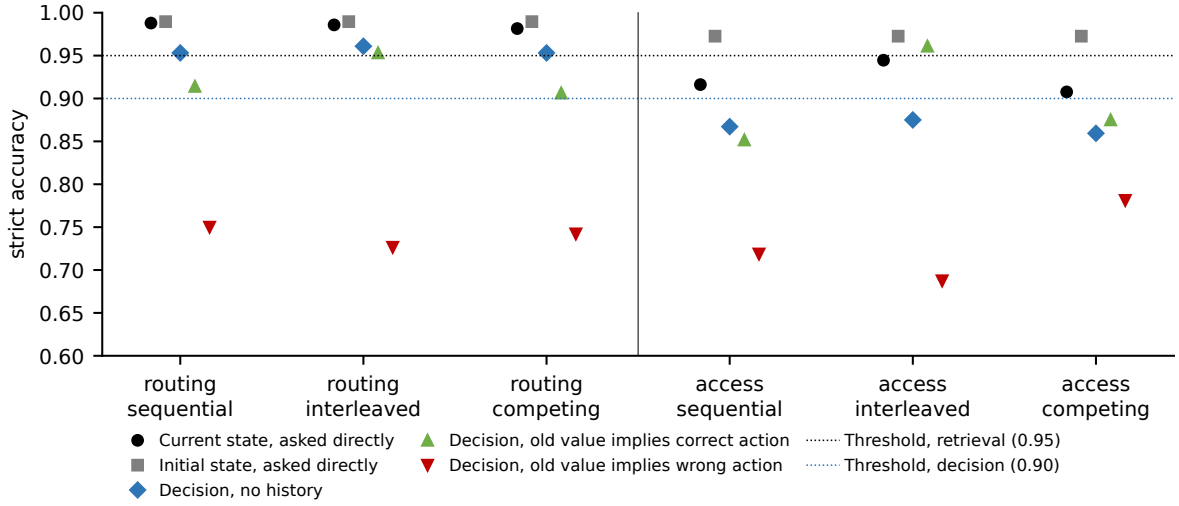


Figure 4: Qwen3-8B, retrieval and decision accuracy per cell on the confirmation records.

0.169 against 0.167, paired difference (mention − superseded) $+0.003$ $[-0.039, +0.044]$, with $\Delta_{\text{SM}} = +0.248$ $[-0.117, 0.601]$; on the histories tracked under all four constructions (Table 12) the difference was -0.003 $[-0.050, 0.042]$. In access (non-competent cells) the superseded assignment moved preference more than the mention ($\Delta_{\text{SM}} = +0.845$ $[0.425, 1.255]$). Easy-query sensitivity did not improve prediction: AUROC 0.684 without and 0.685 with R_{easy}, log-loss gain -0.0001 $[-0.0027, 0.0022]$.

Gate-rule estimates. Competence labels on the confirmation data can differ from the gate labels (Table 10): Qwen3-8B access int. is competent here and not competent on the confirmation data; Qwen3-8B routing comp. is not competent here and competent on the confirmation data; Gemma 4 12B access int. is not competent here and competent on the confirmation data; Gemma 4 12B access seq. is not competent here and competent on the confirmation data; Gemma 4 12B routing int. is competent here and not competent on the confirmation data; Qwen3.5 9B routing comp. is competent here and not competent on the confirmation data; Qwen3.5 9B routing int. is not competent here and competent on the confirmation data; Granite 4.1 8B routing comp. is not competent here and competent on the confirmation data; Granite 4.1 8B routing seq. is competent here and not competent on the confirmation data. Under the amendment's gate-defined cells (Table 10), Qwen3-8B would

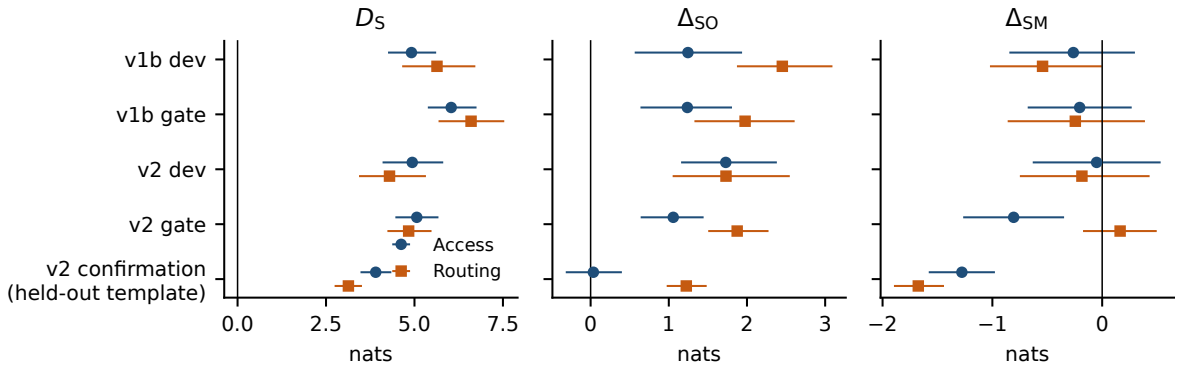


Figure 5: Primary and mention contrasts in the five v1b and v2 splits (97.5% intervals for the confirmation's primary contrasts, 95% otherwise). Only the confirmation is confirmatory.

Stratum	Vocab.	D_S	Δ_{SM}
routing seq.	natural	3.66	−2.32
routing int.	natural	4.03	−1.00
routing comp.	natural	3.07	−2.38
routing seq.	opaque	2.53	−2.17
routing int.	opaque	2.78	−1.32
routing comp.	opaque	2.73	−0.87
access seq.	natural	3.61	−1.16
access int.	natural	4.01	−0.06
access comp.	natural	3.65	−2.21
access seq.	opaque	4.53	−1.10
access int.	opaque	4.43	−1.42
access comp.	opaque	3.20	−1.72

Table 6: Qwen3-8B, effects (nats) by task, cell and vocabulary (64 histories each). Every D_S interval excludes zero; the Δ_{SM} interval includes zero only for access, interleaved, natural ($[-1.01, 0.96]$).

	Overall	Access	Routing
Failures	892/6144	454/3072	438/3072
Log loss, prevalence	0.414	0.419	0.410
Log loss, pre-decision	0.346	0.323	0.358
AUROC, pre-decision	0.780	0.834	0.729
AUROC, $+R_{easy}$	0.779	0.833	0.730
Gain from R_{easy}	+0.0004	−0.0006	+0.0013

Table 7: H3, Qwen3-8B: out-of-fold prediction of strict failure. 95% intervals for the gain: overall $[-0.0011, 0.0019]$, access $[-0.0020, 0.0008]$, routing $[-0.0022, 0.0047]$.

use routing int.+seq. ($n = 256$): $D_S = +3.248$, $\Delta_{SO} = +1.330$, net switch rate $+0.195$; access int. ($n = 128$): $D_S = +4.221$, $\Delta_{SO} = -0.132$, net switch rate $+0.273$. Qwen3-14B's cells are unchanged. Gemma 4 12B would use routing comp.+int.+seq. ($n = 384$): $D_S = +3.949$, $\Delta_{SO} = +3.247$, net switch rate $+0.148$; access comp. ($n = 128$): $D_S = +4.384$, $\Delta_{SO} = +1.184$, net switch rate $+0.062$. Qwen3.5 9B would use routing comp.+seq. ($n = 256$): $D_S = +1.085$, $\Delta_{SO} = +0.358$, net switch rate $+0.230$. Granite 4.1 8B would use routing int.+seq. ($n = 256$): $D_S = +3.441$, $\Delta_{SO} = +0.793$, net switch rate $+0.168$.

C Reproducibility

Models. Qwen3-8B, Hugging Face revision `b968826d9c46dd6066d109eabc6255188de91218`; Qwen3-14B, revision `40c069824f4251a91eefaf281ebe4c544efd3e18`; Gemma 3 4B (instruction-tuned), revision `093f9f388b31de276ce2de164bdc2081324b9767`; Ministral 3 14B (instruction-tuned, BF16 repository), revision `3cea74c1ebaf5ce5f5a2553de470e2ceab825142`; Gemma 4 12B (instruction-tuned), revision `707f0a3b8a3c7ad586ed01e27eafbad8a27dd0f7`; Qwen3.5 9B, revision `c202236235762e1c871ad0ccb60c8ee5ba337b9a`; Granite 4.1 8B, revision `1504002f650e656a0a3789d99574df12e3e94ed0`. All ran in float32 without quantization, with SDPA attention, on one 80 GB H100 GPU, using Transformers (Wolf et al., 2020) and PyTorch (Paszke et al., 2019). Licenses: Gemma 3 (Gemma Terms of Use) and Apache 2.0 for all other models, used for non-commercial research within their terms.

Artifacts. Each dataset is regenerated deterministically from its seed and checked against its manifest. Each score file carries per-record checksums and a completion record binding the scores, retrieval scores, run manifest,

	Main design, no thinking ($n = 384$)			Balanced, no thinking ($n = 192$)			Main design, thinking ($n = 384$)	
Construction	D	Net switch	W/C	D	Net switch	W/C	Net switch	W/C
Superseded	$+3.133$ $[+2.794, +3.478]$	$+0.185$ $[+0.143, +0.227]$	75/4	$+3.895$ $[+3.333, +4.443]$	$+0.031$ $[+0.010, +0.057]$	6/0	$+0.000$ $[+0.000, +0.000]$	0/0
Other attr., updated	$+1.909$ $[+1.661, +2.161]$	$+0.065$ $[+0.029, +0.099]$	37/12	$+2.962$ $[+2.541, +3.365]$	$+0.026$ $[+0.005, +0.052]$	5/0	$+0.000$ $[+0.000, +0.000]$	0/0
Mention	$+4.807$ $[+4.455, +5.186]$	$+0.294$ $[+0.250, +0.341]$	113/0	$+6.621$ $[+5.917, +7.323]$	$+0.177$ $[+0.125, +0.234]$	34/0	$+0.018$ $[+0.005, +0.034]$	7/0
Unattached	$+2.496$ $[+2.201, +2.805]$	$+0.039$ $[+0.013, +0.068]$	23/8	$+3.326$ $[+2.889, +3.765]$	-0.005 $[-0.026, +0.010]$	1/2	$+0.000$ $[+0.000, +0.000]$	0/0

Table 8: Values behind Figure 2 (Qwen3-8B, routing), with 95% intervals. W/C: histories switched to the wrong / to the correct action.

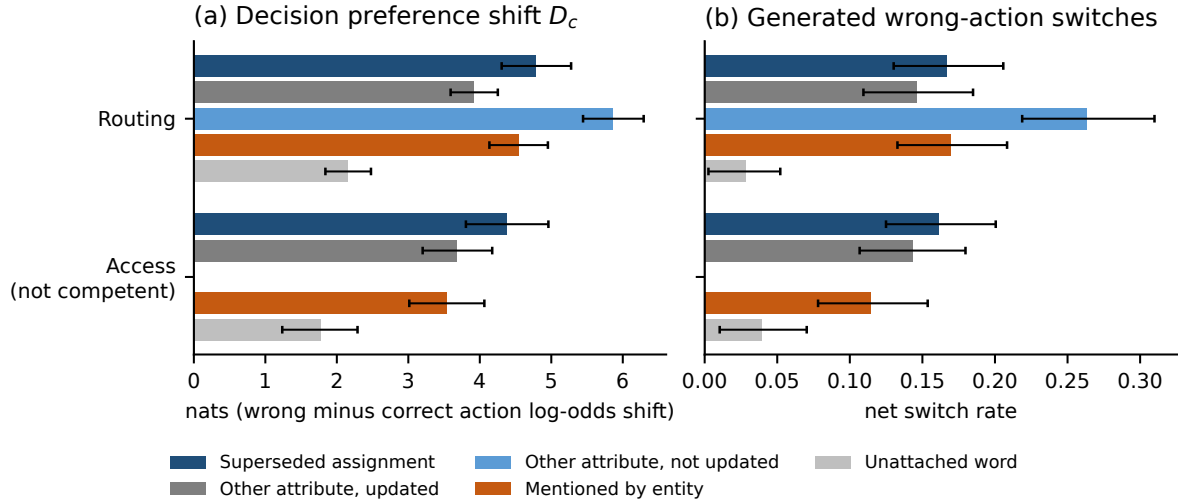


Figure 6: Qwen3-14B, as Figure 3.

	Qwen3-8B		Qwen3-14B	
	D	Net switch	D	Net switch
Superseded	$+3.133$ $[+2.794, +3.478]$	$+0.185$ $[+0.143, +0.227]$	$+4.785$ $[+4.306, +5.277]$	$+0.167$ $[+0.130, +0.206]$
Other attr., updated	$+1.909$ $[+1.661, +2.161]$	$+0.065$ $[+0.029, +0.099]$	$+3.919$ $[+3.592, +4.252]$	$+0.146$ $[+0.109, +0.185]$
Other attr., not updated	$+3.112$ $[+2.815, +3.397]$	$+0.193$ $[+0.154, +0.234]$	$+5.861$ $[+5.444, +6.292]$	$+0.263$ $[+0.219, +0.310]$
Mention	$+4.807$ $[+4.455, +5.186]$	$+0.294$ $[+0.250, +0.341]$	$+4.538$ $[+4.134, +4.952]$	$+0.169$ $[+0.133, +0.208]$
Unattached	$+2.496$ $[+2.201, +2.805]$	$+0.039$ $[+0.013, +0.068]$	$+2.145$ $[+1.840, +2.477]$	$+0.029$ $[+0.003, +0.052]$
Δ_{ON} (updated − not updated)	-1.204 $[-1.326, -1.080]$	-0.128 $[-0.177, -0.081]$	-1.942 $[-2.191, -1.706]$	-0.117 $[-0.167, -0.070]$
Δ_{SO} (superseded − updated)	$+1.224$ $[+1.002, +1.454]$	$+0.120$ $[+0.076, +0.164]$	$+0.866$ $[+0.476, +1.263]$	$+0.021$ $[-0.023, +0.068]$
Δ_{SN} (superseded − not updated)	$+0.020$ $[-0.201, +0.254]$	-0.008 $[-0.052, +0.036]$	-1.076 $[-1.526, -0.661]$	-0.096 $[-0.154, -0.044]$

Table 9: Never-updated other attribute (N) on the routing confirmation histories ($n = 384$), with 95% intervals. Prediction recorded before scoring: $\Delta_{ON} \leq 0$.

tokenizer audit, code hash and configuration hash. Analyses refuse incomplete or altered runs. Confirmation: seed 82003, dataset hash `3e972257...`, code hash `c6648aba...`, configuration hash `78d09215...`, score file SHA-256 `b159b727....` The protocol was frozen on 2026-10-08 (UTC), before the gate and confirmation splits were generated. Every number and figure in this paper is produced from the sealed analysis reports by one script, which records the hash of each input.

Compute. Qwen3-8B's confirmation scoring took 6.27 GPU hours (2.67 s per record, including two retrieval queries, one historical query where applicable, two scored action events and greedy generation). Qwen3-14B took 1.12 (development), 2.23 (gate) and 4.48 GPU hours (confirmation), 1.91 s per confirmation record. Both runs used the same GPU type, software versions (PyTorch 2.14.0, Transformers 5.17.0), scorer and prompts (mean 199 prompt tokens). The times are wall-clock as logged. We did not record GPU sharing or clock state, so we cannot attribute the faster per-record time of the larger model. Confirmation scoring took 4.29 (Gemma 4 12B), 7.59 (Qwen3.5 9B) and 2.67 GPU hours (Granite 4.1 8B); Qwen3.5 ran its recurrent layers through the library's reference PyTorch path, without fused kernels. Generation with thinking enabled took 6.51 (Qwen3-8B) and 8.77 GPU hours (Qwen3-14B). The logged time of all 33 timed runs, including the v1b splits, development and gate splits, the ineligible models, the never-updated attribute, the balanced subset and thinking mode, sums to 77.8 GPU hours. Preflight and numerical checks, model downloads and runs stopped before completion were not timed and add to this.

Code. The anonymized archive submitted as supplementary software contains the generators, scoring and analysis code, tests, configuration files, the dated protocol records, the sealed analysis reports with their run records, the confirmation datasets and per-record scores of the five confirmed mod-

Model	Cell	Direct	No hist.	History	Comp.
Qwen3-8B	routing seq.	0.977	0.938	0.779	yes
	routing int.	0.986	0.938	0.779	yes
	routing comp.	0.945	0.938	0.752	no
	access seq.	0.889	0.906	0.822	no
	access int.	0.953	0.938	0.877	yes
	access comp.	0.953	0.859	0.883	no
Qwen3-14B	routing seq.	0.996	0.922	0.854	yes
	routing int.	1.000	1.000	0.846	yes
	routing comp.	0.997	1.000	0.814	yes
	access seq.	0.991	0.781	0.729	no
	access int.	0.977	0.828	0.748	no
	access comp.	0.946	0.828	0.697	no
Gemma 3 4B	routing seq.	0.939	0.688	0.645	no
	routing int.	0.964	0.609	0.488	no
	routing comp.	0.916	0.625	0.533	no
	access seq.	0.945	0.719	0.678	no
	access int.	0.936	0.609	0.676	no
	access comp.	0.946	0.672	0.641	no
Ministral 3 14B	routing seq.	0.997	0.766	0.766	no
	routing int.	0.986	0.766	0.756	no
	routing comp.	0.999	0.812	0.756	no
	access seq.	0.970	0.750	0.670	no
	access int.	0.926	0.812	0.711	no
	access comp.	0.952	0.797	0.732	no
Gemma 4 12B	routing seq.	1.000	0.906	0.936	yes
	routing int.	0.999	0.984	0.887	yes
	routing comp.	1.000	0.969	0.930	yes
	access seq.	0.912	0.969	0.967	no
	access int.	0.987	0.875	0.850	no
	access comp.	0.993	0.906	0.932	yes
Qwen3.5 9B	routing seq.	1.000	0.906	0.795	yes
	routing int.	0.922	0.922	0.746	no
	routing comp.	0.950	0.969	0.867	yes
	access seq.	0.994	0.688	0.656	no
	access int.	0.908	0.703	0.666	no
	access comp.	0.891	0.688	0.789	no
Granite 4.1 8B	routing seq.	0.967	0.984	0.799	yes
	routing int.	0.955	0.953	0.738	yes
	routing comp.	0.932	0.953	0.816	no
	access seq.	0.935	0.797	0.828	no
	access int.	0.960	0.812	0.793	no
	access comp.	0.962	0.828	0.865	no

Table 10: Frozen gate split (384 histories, the same rows for every model): accuracy and competence per cell. This gate decided eligibility for confirmation.

els, and the scripts that derive every number, table and figure in this paper; run from the archive, they reproduce the paper's derived values exactly.

Model	Construction	Kept	Net switch rate	Switched
Qwen3-8B	superseded	348/384	$+0.158$ $[+0.118, +0.198]$	58 / 3
	other attr.	357/384	$+0.048$ $[+0.014, +0.084]$	29 / 12
	mention	349/384	$+0.278$ $[+0.232, +0.327]$	97 / 0
	unattached	351/384	$+0.034$ $[+0.009, +0.060]$	17 / 5
Qwen3-14B	superseded	376/384	$+0.165$ $[+0.128, +0.205]$	65 / 3
	other attr.	367/384	$+0.136$ $[+0.101, +0.174]$	52 / 2
	mention	363/384	$+0.168$ $[+0.132, +0.207]$	61 / 0
	unattached	375/384	$+0.024$ $[+0.000, +0.051]$	16 / 7
Gemma 4 12B	superseded	349/384	$+0.143$ $[+0.109, +0.181]$	50 / 0
	other attr.	349/384	$+0.006$ $[-0.012, +0.026]$	7 / 5
	mention	339/384	$+0.068$ $[+0.041, +0.094]$	23 / 0
	unattached	343/384	$+0.009$ $[-0.015, +0.032]$	10 / 7
Qwen3.5 9B	superseded	326/384	$+0.233$ $[+0.184, +0.279]$	77 / 1
	other attr.	328/384	$+0.085$ $[+0.058, +0.116]$	28 / 0
	mention	314/384	$+0.191$ $[+0.150, +0.236]$	60 / 0
	unattached	345/384	$+0.038$ $[+0.014, +0.064]$	17 / 4
Granite 4.1 8B	superseded	331/384	$+0.145$ $[+0.109, +0.184]$	48 / 0
	other attr.	340/384	$+0.124$ $[+0.091, +0.159]$	42 / 0
	mention	310/384	$+0.326$ $[+0.274, +0.377]$	101 / 0
	unattached	326/384	$+0.175$ $[+0.135, +0.218]$	58 / 1

Table 11: Routing histories each model tracks (criteria of §5.2), with 95% intervals. Retained sets differ by construction; Table 12 uses a common set.

	Qwen3-8B $n = 335/384$	Qwen3-14B $n = 359/384$	Gemma 4 12B $n = 339/384$	Qwen3.5 9B $n = 294/384$	Granite 4.1 8B $n = 299/384$
S	$+0.161$ $[+0.119, +0.203]$	$+0.170$ $[+0.128, +0.212]$	$+0.147$ $[+0.109, +0.186]$	$+0.228$ $[+0.180, +0.279]$	$+0.157$ $[+0.117, +0.201]$
O	$+0.045$ $[+0.009, +0.081]$	$+0.139$ $[+0.103, +0.178]$	$+0.006$ $[-0.015, +0.027]$	$+0.085$ $[+0.054, +0.119]$	$+0.127$ $[+0.090, +0.164]$
M	$+0.266$ $[+0.221, +0.313]$	$+0.167$ $[+0.131, +0.209]$	$+0.068$ $[+0.041, +0.094]$	$+0.177$ $[+0.136, +0.221]$	$+0.321$ $[+0.271, +0.371]$
U	$+0.033$ $[+0.006, +0.060]$	$+0.025$ $[+0.000, +0.053]$	$+0.009$ $[-0.015, +0.032]$	$+0.044$ $[+0.017, +0.075]$	$+0.184$ $[+0.140, +0.227]$
M − S	$+0.104$ $[+0.057, +0.155]$	-0.003 $[-0.050, +0.042]$	-0.080 $[-0.124, -0.038]$	-0.051 $[-0.102, +0.000]$	$+0.164$ $[+0.117, +0.217]$
S − O	$+0.116$ $[+0.075, +0.164]$	$+0.031$ $[-0.014, +0.078]$	$+0.142$ $[+0.103, +0.189]$	$+0.143$ $[+0.095, +0.190]$	$+0.030$ $[-0.007, +0.067]$
O − U	$+0.012$ $[-0.024, +0.051]$	$+0.114$ $[+0.070, +0.159]$	-0.003 $[-0.029, +0.024]$	$+0.041$ $[+0.003, +0.082]$	-0.057 $[-0.104, -0.010]$

Table 12: Net switch rates on the routing histories tracked under all four constructions (post hoc), with 95% intervals; lower rows are paired differences. S, O, M, U: §3.

			Scores (nats)			Net switch rate		
Model	Cells	n	D_{S}	Δ_{SO}	Δ_{SM}	S	S − O	M − S
Qwen3-8B	routing comp., int., seq.	384	$+3.133$ $[+2.743, +3.516]$	$+1.224$ $[+0.973, +1.485]$	-1.675 $[-1.896, -1.441]$	$+0.185$ $[+0.143, +0.227]$	$+0.120$ $[+0.076, +0.164]$	$+0.109$ $[+0.060, +0.156]$
Qwen3-14B	routing comp., int., seq.	384	$+4.785$ $[+4.244, +5.354]$	$+0.866$ $[+0.417, +1.328]$	$+0.248$ $[-0.117, +0.601]$	$+0.167$ $[+0.130, +0.206]$	$+0.021$ $[-0.023, +0.068]$	$+0.003$ $[-0.039, +0.044]$
Gemma 4 12B	routing comp., seq.	256	$+3.709$ $[+3.146, +4.247]$	$+3.102$ $[+2.564, +3.642]$	$+1.513$ $[+0.998, +2.041]$	$+0.109$ $[+0.070, +0.148]$	$+0.090$ $[+0.051, +0.133]$	-0.012 $[-0.062, +0.039]$
Gemma 4 12B	access comp., int., seq.	384	$+4.285$ $[+3.622, +4.998]$	$+1.561$ $[+0.743, +2.345]$	$+1.455$ $[+0.823, +2.043]$	$+0.083$ $[+0.057, +0.112]$	-0.031 $[-0.076, +0.008]$	-0.010 $[-0.042, +0.026]$
Qwen3.5 9B	routing int., seq.	256	$+1.072$ $[+0.939, +1.209]$	$+0.311$ $[+0.228, +0.399]$	-0.019 $[-0.079, +0.035]$	$+0.242$ $[+0.191, +0.297]$	$+0.148$ $[+0.098, +0.199]$	-0.062 $[-0.125, +0.000]$
Granite 4.1 8B	routing comp., int.	256	$+3.385$ $[+2.851, +3.957]$	$+0.755$ $[+0.483, +1.053]$	-1.929 $[-2.365, -1.512]$	$+0.188$ $[+0.141, +0.238]$	$+0.027$ $[-0.012, +0.062]$	$+0.156$ $[+0.098, +0.215]$

Table 13: Table 3 with intervals: 97.5% for D_{S} and Δ_{SO}, 95% otherwise (history bootstrap).